

HV-SESAME Analyzer

(V1)

USER GUIDE

H/V Spectral Ratio SESAME Reliability and Clear-Peak Assessment Tool

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1. Purpose and Scope

HV-SESAME Analyzer V1 is a graphical post-processing application for batch evaluation of H/V spectral-ratio results using the implemented SESAME reliability and clear-peak criteria. The application reads paired Geopsy .hv and .log files, evaluates criteria V1–V9, displays station-level H/V curves and criterion results, assigns a project-level quality class, and produces reporting and spatial-export products.

The software is intended to be used after H/V processing has been completed in Geopsy. It does not replace the original signal-processing stage; instead, it provides an integrated quality-control, visualization, reporting, mapping, and project-management workflow.

2. Required Input Files

2.1 Geopsy H/V files

All .hv files and their corresponding .log files must be stored in the same data folder. The two files belonging to one station must have the same base filename.

Example:

m1.hv-m1.log
m2.hv-m2.log
m10.hv -m10.log

The analyzer scans the selected folder for .hv files. If the matching .log file is not found, that station is skipped. Station names are naturally sorted so that names such as m1, m2, ..., m10 are presented in numerical rather than purely alphabetical order.

2.2 Coordinate file

A coordinate file is required in the current V1 workflow. The file must be a TXT or DAT file containing four fields in the order X, Y, Z, Name. Tab-separated input is recommended.

X	Y	Z	Name
450123.55	4578123.21	105.3	m1
450486.20	4577950.40	112.8	m2

The Name field must correspond to the base name of the Geopsy files. For example, coordinate name m1 is associated with m1.hv and m1.log.

2.3 UTM zone

Coordinates are interpreted using WGS 84 UTM. The default value shown by the application is 35N. Before running the analysis, verify that the displayed zone is correct for the survey area and edit it when necessary.

3. Starting the Application

Run HV_SESAME_ANALYZER_V1 in MATLAB. The main window opens with a workflow bar at the top, a station panel on the left, an H/V spectral-ratio graph in the center, a criteria/summary panel on the right, and a status field at the bottom (Fig 1).

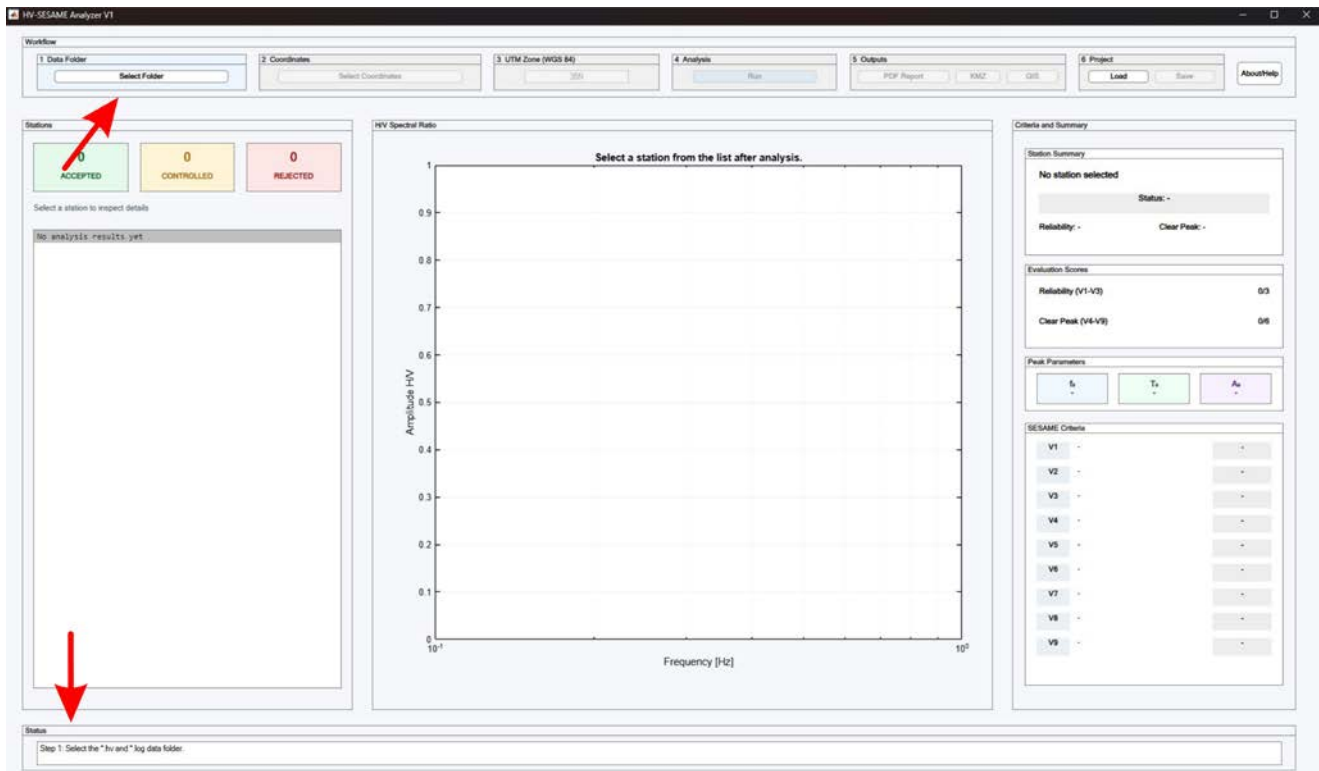


Figure 1. HV-SESAME Analyzer main graphical user interface.

The workflow is intentionally sequential. Buttons become available as the required preceding steps are completed. A previously saved project can also be opened directly with Load.

4. Step-by-Step Analysis Workflow

Step 1 — Select the H/V data folder

Click Select Folder in the 1 Data Folder panel. Choose the directory containing the paired .hv and .log files. After a valid folder is selected, the coordinate-selection stage becomes active. The status field confirms completion of Step 1 (Fig 2).

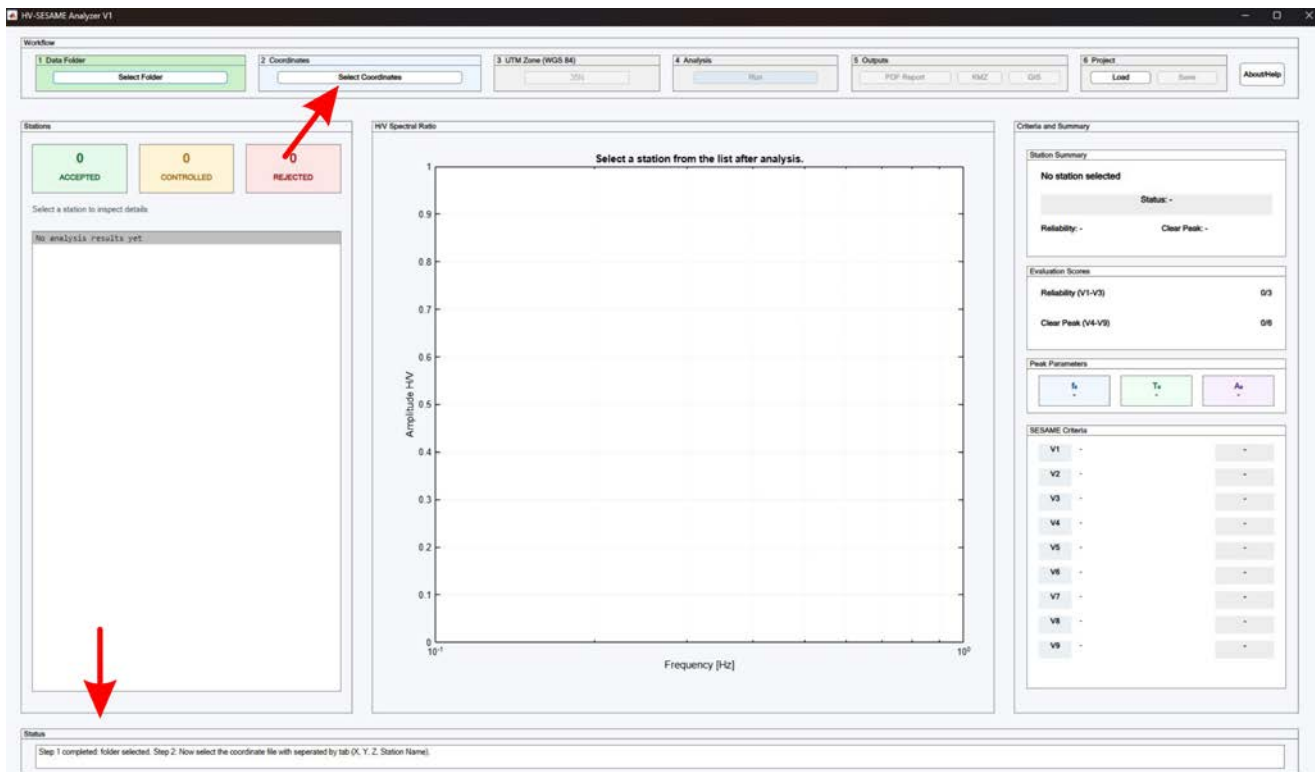


Figure 2. HV-SESAME Analyzer V1 after selection of the H/V data folder. The coordinate-file selection stage becomes active, and the status bar guides the user to the next step.

Step 2 — Import station coordinates

Click Select Coordinates and choose the TXT or DAT coordinate file. The application reads the X, Y, Z and Name fields. If the file cannot be interpreted, a Coordinate Reading Error dialog is displayed. A successful import enables the UTM-zone field and Run button (Fig 3).

Step 3 — Check the WGS 84 UTM zone

Inspect the value in UTM Zone (WGS 84). The initial value is 35N. Replace it with the correct zone for the project when required. This value is used for spatial conversion and export (Fig 3).

Step 4 — Run the analysis

Click Run. The analyzer scans all .hv files, locates their matching .log files, evaluates the available station pairs, constructs the result tables, and prepares latitude/longitude information for spatial outputs. During processing the status field displays Analysis is running.... When the batch is complete, it reports the total number of analyzed stations (Fig 3).

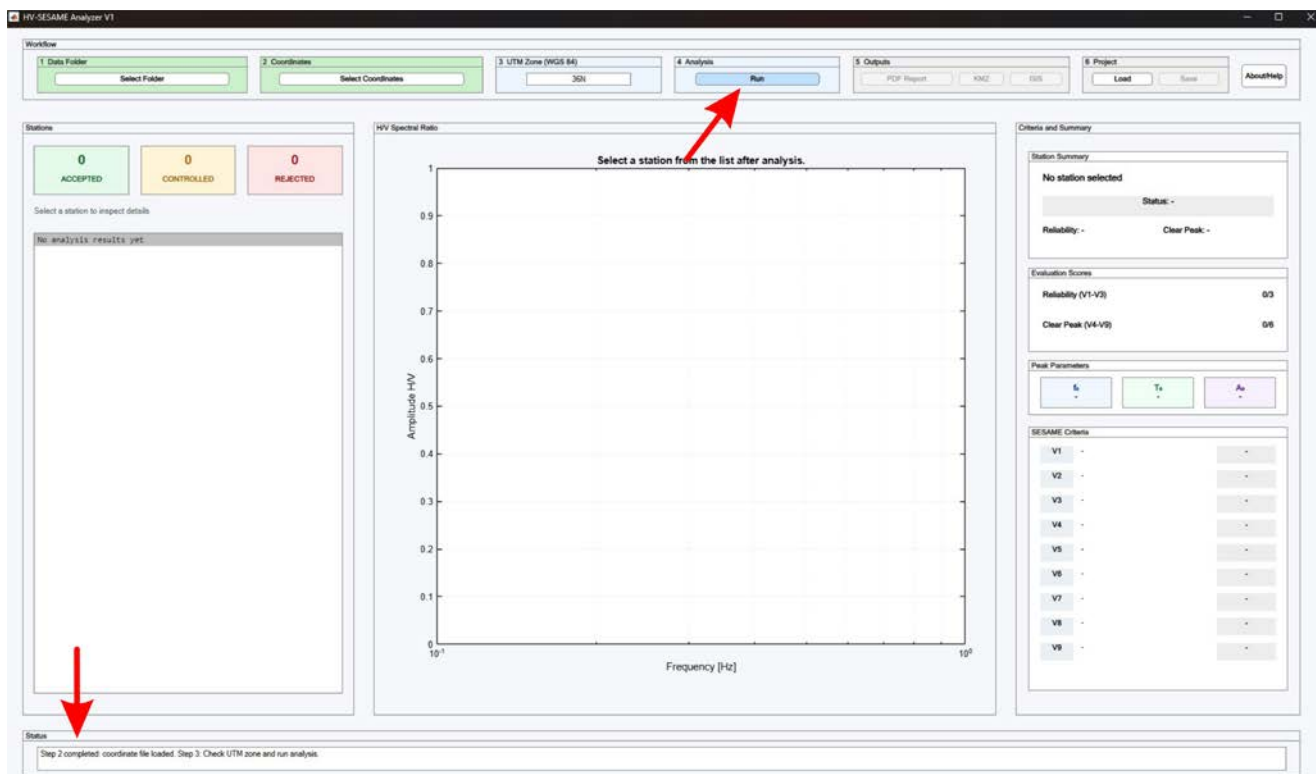


Figure 3. Interface after importing the station-coordinate file. The user verifies the WGS 84 UTM zone and starts the automated batch analysis by clicking Run

5. Reading the Main Results Window

5.1 Station classification panel

After analysis, the left panel reports the number of stations in the three quality classes: Accepted, Controlled, and Rejected. The station list below these counters is used to inspect individual results.

5.2 Selecting a station

Click a station in the list. The central and right-hand panels are updated for the selected station. This allows rapid review of all measurements without reopening the original files.

5.3 H/V spectral-ratio plot

The central H/V Spectral Ratio panel displays the selected station's H/V curve on a logarithmic frequency axis. Frequency is shown in Hz and the vertical axis represents H/V amplitude.

5.4 Station Summary

The Station Summary panel shows the station name, overall classification, Reliability result, and Clear Peak result. The Evaluation Scores section summarizes the number of passed criteria within the two criterion groups.

Evaluation group	Criteria	Displayed score
Reliability	V1-V3	0/3 to 3/3
Clear Peak	V4-V9	0/6 to 6/6

5.5 Peak parameters

Three parameter cards display f_0 , T_0 and A_0 for the selected H/V peak. These provide the frequency, corresponding period, and H/V amplitude associated with the identified peak.

5.6 SESAME criteria V1–V9

The SESAME Criteria panel lists V1 through V9 together with a short criterion description and its TRUE/FALSE evaluation. V1–V3 form the Reliability group, while V4–V9 form the Clear Peak group.

6. Quality Classes

The analyzer uses three project-level display classes. Accepted stations are shown in green, Controlled Acceptance stations in yellow, and Rejected stations in red. These same classes are used consistently in the interface and spatial/reporting products.

Criterion-level TRUE/FALSE results should remain the primary basis for technical interpretation. The three-class summary is intended to make project-scale screening and mapping easier.

7. Exporting Results

7.1 PDF Report

After a successful analysis, click PDF Report and select the destination filename. The report contains project-level summary material and analysis graphics/tables generated by the software. The implementation produces report pages using MATLAB print-based output (Fig 4).

For multi-page output, the software creates the pages individually. If Ghostscript is available, the pages are automatically merged into a single PDF. If Ghostscript is not available, the individual PDF pages are retained instead.

7.2 KMZ export

Click KMZ and choose an output filename. The spatial results are written as a Google Earth KMZ product using the geographic coordinates generated from the imported UTM information. The classification convention uses green, yellow, and red for Accepted, Controlled Acceptance, and Rejected stations(Fig 4).

7.3 GIS export

Click GIS to create GIS-compatible output. Before saving, the application asks which stations should be included. Accepted + Controlled is the recommended default; All Stations also includes rejected measurements in the exported attribute data.

Choose the destination .shp filename. The GIS export generates the shapefile components required for use in GIS software and also writes the associated tabular CSV output. Keep the shapefile component files together when moving or sharing the dataset(Fig 4).

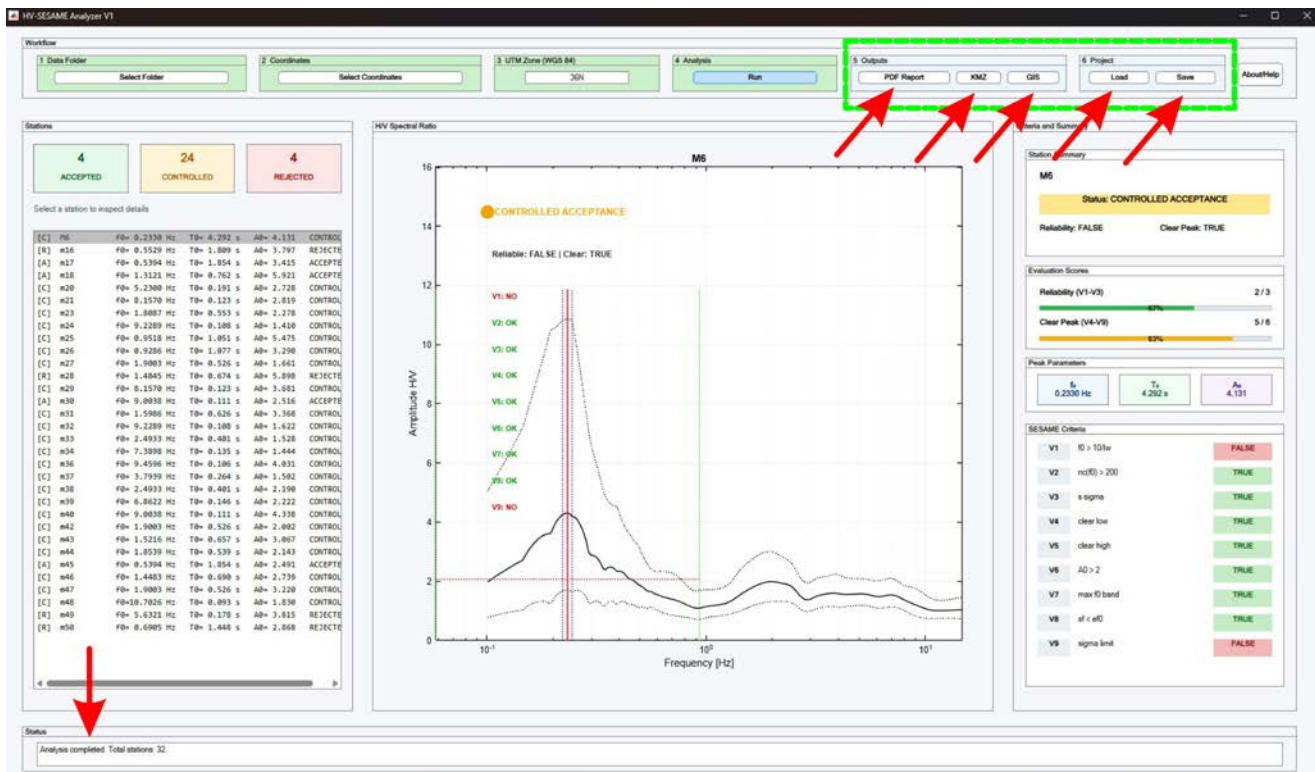


Figure 4. Main results interface after completion of the automated analysis. The interface provides project-level quality-class counts, station selection, the H/V spectral-ratio curve, Reliability and Clear-Peak scores, peak parameters, individual SESAME criterion results (V1-V9), and access to PDF, KMZ, GIS, Save, and Load functions.

8. Saving and Reopening a Project

8.1 Save

After an analysis has been completed or loaded, click Save. Projects can be stored as .hsap or .mat files. The project structure preserves the data-folder reference, coordinate-file reference, UTM zone, coordinate table, station results, final result table, and geographic-coordinate table (Fig 4).

8.2 Load

Click Load to open an existing .hsap or .mat project. A valid project restores the analysis results and interface state, including the UTM zone and station list, so that stations can be inspected and outputs can be generated again without rerunning the original batch analysis (Fig 4).

9. Recommended Working Procedure

1. Complete the H/V processing in Geopsy and export matching .hv and .log files.
2. Place all paired station files for the project in one folder.
3. Prepare the X-Y-Z-Name coordinate TXT/DAT file and verify that station names match the file base names.
4. Start HV-SESAME Analyzer V1 and select the data folder.
5. Load the coordinate file.

6. Verify the WGS 84 UTM zone.
7. Run the batch analysis.
8. Inspect the Accepted, Controlled, and Rejected station counts.
9. Review individual H/V curves, peak parameters, and V1–V9 results.
10. Export the PDF, KMZ, and/or GIS products required for the project.
11. Save an .hsap project file if the analysis will be revisited later.

10. Common Problems and Checks

Problem	Check
Run button is disabled	Confirm that both the data folder and coordinate file have been selected successfully.
A station is missing from the results	Check that the .hv file has a matching .log file with exactly the same base name.
Coordinate import fails	Check the X Y Z Name structure, delimiter consistency, and station-name field.
Spatial output is misplaced	Verify the WGS 84 UTM zone and the imported X/Y coordinate values.
PDF is not merged into one file	Ghostscript may not be available; the software can retain the report as separate PDF pages.

11. About / Help

The About/Help button displays the software name, its purpose, available output types, status-color convention, and developer contact information.

12. Important Use Note

HV-SESAME Analyzer V1 is a post-processing and quality-assessment tool. Users should retain the original Geopsy outputs and evaluate the H/V results in their geological, geophysical, and site-specific context. Automated classification facilitates reproducible screening but does not remove the need for professional interpretation.